Project Sentinel

UPI Transaction Data Engineering Pipeline using Databricks

1. Project Overview

Project Sentinel is an end-to-end Data Engineering project that implements the Medallion Architecture (Landing, Bronze, Silver and Gold) using Databricks. The project generates realistic UPI transaction data, performs ETL processing, builds business-ready summary tables, and detects suspicious transactions using rule-based fraud detection. As per mentor guidance, static CSV batch data was used instead of JSON streaming data.

2. Objectives

Generate realistic UPI transaction data, implement an ETL pipeline, improve data quality, create Gold summary tables, detect suspicious transactions, and validate the complete pipeline.

3. Technologies Used

Databricks, Apache Spark (PySpark), Delta Lake, Python, Pandas, SQL, Git and GitHub.

4. Project Workflow

Notebook 1 generates the CSV dataset. Notebook 2 loads the Landing data into the Bronze layer. Notebook 3 cleans and standardizes the Bronze data into the Silver layer. Notebook 4 creates Gold summary tables. Notebook 5 performs rule-based fraud detection. Notebook 6 validates the pipeline and generates business KPIs and a data quality report.

5. Medallion Architecture

Landing stores raw CSV data. Bronze stores raw Delta data with metadata. Silver contains cleaned and standardized data. Gold contains business-ready summary tables for reporting and dashboards.

6. Fraud Detection

Implemented three business rules: High Value Transactions, High Response Time, and Multiple Failed Transactions by the same sender. Suspicious records are stored in a separate fraud table.

7. Data Quality

Performed duplicate removal, null handling, trimming of extra spaces, data type correction, removal of invalid negative transactions, standardization of text fields, and masking of sensitive UPI IDs.

8. Business Outputs

Created Daily Summary, Bank Summary, City Summary, and Transaction Type Summary tables for reporting and analytics.

9. Learning Outcomes

Hands-on experience with Databricks, Spark, Delta Lake, ETL pipelines, Medallion Architecture, Data Quality, Fraud Detection, and GitHub documentation.

10. Conclusion

Project Sentinel demonstrates a complete batch data engineering pipeline for UPI transaction analysis using Databricks. The project showcases ETL development, data quality improvement, business analytics, and rule-based fraud detection in a structured and professional workflow.